LaTeX – Typesetting Mathematical Formulas

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UP FAMNIT

Single Equations

Math Packages

For typesetting mathematical formulas load **amsmath, amsfonts** and **amssymb** packages in the preamble:

```
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amssymb}
```

Text Style vs. Display Style

A mathematical formula can be typeset:

- in-line within a paragraph (*text style*), or
- the paragraph can be broken and the formula typeset separately (*display style*).

Text Style

Mathematical equations *within* a paragraph (*text style*) are entered between $ and $:

```
Add $a$ squared and $b$ squared to get
$c$ squared. Or, using a more
mathematical approach: $a^2 + b^2 = c^2$.
```

Add a squared and b squared to get c squared. Or, using a more mathematical approach: $a^2 + b^2 = c^2$.

Display Style - Equation Environment

- If you want your equations to be set apart from the rest of the paragraph (*display style*) enclose them between

  `\begin{equation}` and `\end{equation}`.

- You can then `\label` an equation number and refer to it somewhere else in the text by using the `\eqref` command.
- If you want to name the equation something specific, you `\tag` it instead.

Display Style - Example

```
Add $a$ squared and $b$ squared to get $c$ squared.
Or, using a more mathematical approach
\begin{equation}
  a^2 + b^2 = c^2
\end{equation}
Einstein says
\begin{equation}
  E = mc^2 \label{clever}
\end{equation}
He did not say
\begin{equation}
  1 + 1 = 3 \tag{dumb}
\end{equation}
This is a reference to
\eqref{clever}.
```

Display Style - Example

Add a squared and b squared to get c squared. Or, using a more mathematical approach

$$a^2 + b^2 = c^2 \tag{1}$$

Einstein says

$$E = mc^2 \tag{2}$$

He did not say

$$1 + 1 = 3 \tag{dumb}$$

This is a reference to (2).

Display Style

If you **don't want LaTeX to number** the equations:

- use the starred version of `equation` using an asterisk, **equation***, or
- enclose the equation in `$$` and `$$`, or
- enclose the equation in `\[` and `\]`.

Display Style - Example

```
Add $a$ squared and $b$ squared to
get $c$ squared.
Or, using a more mathematical approach
\begin{equation*}
a^2 + b^2 = c^2
\end{equation*}
or you can type less for the same effect:
$$a^2 + b^2 = c^2$$
\[ a^2 + b^2 = c^2 \]
```

Display Style - Example

Add a squared and b squared to get c squared. Or, using a more mathematical approach

$$a^2 + b^2 = c^2$$

or you can type less for the same effect:

$$a^2 + b^2 = c^2$$

$$a^2 + b^2 = c^2$$

Text Style vs. Display Style

The difference in typesetting style between *text style* and *display style* equations:

```
This is text style:
$\lim_{n \to \infty}\sum_{k=1}^n \frac{1}{k^2} =
\frac{\pi^2}{6}$.
And this is display style:
\begin{equation}
\lim_{n \to \infty}\sum_{k=1}^n \frac{1}{k^2} =
\frac{\pi^2}{6}
\end{equation}
```

Text Style vs. Display Style

This is text style: $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{k^2} = \frac{\pi^2}{6}$.

And this is display style:

$$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{k^2} = \frac{\pi^2}{6} \tag{3}$$

Math Mode

In *math mode* (text style or display style):

- spaces have to be specified with special commands such as `\,`, `\quad` or `\qquad`;
- empty lines are not allowed;
- each letter is considered to be the name of a variable and will be typeset as such;
- if you want to typeset normal text within a formula (normal upright font and normal spacing) then you have to enter the text using the `\text{...}` command.

Building Blocks of a Mathematical Formula

Greek Letters

- Lowercase Greek letters are entered as `\alpha`, `\beta`, `\gamma`, ...
- Uppercase Greek letters are entered as `\Gamma`, `\Delta`, ...

```
$\lambda,\xi,\pi,\theta,\mu,\Phi,\Omega$
```

$\lambda, \xi, \pi, \theta, \mu, \Phi, \Omega$

Greek Letters

There is no uppercase of some of the letters like `\Alpha`, `\Beta` and so on, because they look the same as normal roman letters: A, B...

α	\alpha	θ	\theta	o	o	υ	\upsilon
β	\beta	ϑ	\vartheta	π	\pi	ϕ	\phi
γ	\gamma	ι	\iota	ϖ	\varpi	φ	\varphi
δ	\delta	κ	\kappa	ρ	\rho	χ	\chi
ϵ	\epsilon	λ	\lambda	ϱ	\varrho	ψ	\psi
ε	\varepsilon	μ	\mu	σ	\sigma	ω	\omega
ζ	\zeta	ν	\nu	ς	\varsigma		
η	\eta	ξ	\xi	τ	\tau		
Γ	\Gamma	Λ	\Lambda	Σ	\Sigma	Ψ	\Psi
Δ	\Delta	Ξ	\Xi	Υ	\Upsilon	Ω	\Omega
Θ	\Theta	Π	\Pi	Φ	\Phi		

Exponents, Superscripts and Subscripts

- Exponents, Superscripts and Subscripts can be specified using the `^` and the `_` characters.
- Most math mode commands act only on the next character, so if you want a command to affect several characters, you have to group them together using curly braces: `{...}`.

Exponents, Superscripts and Subscripts - Examples

```
$p^3_{ij} \qquad
m_\text{Knuth}\qquad
\sum_{k=1}^3 k$ \\[5pt]
$a^x+y \neq a^{x+y}\qquad
e^{x^2} \neq {e^x}^2$
```

$p^3_{ij} \qquad m_\text{Knuth}\qquad \sum_{k=1}^3 k$

$a^x+y \neq a^{x+y}\qquad e^{x^2} \neq {e^x}^2$

Binary Relations

You can negate the following symbols by prefixing them with a `\not` command.

$<$	`<`	$>$	`>`	$=$	`=`
$\leq$	`\leq` or `\le`	$\geq$	`\geq` or `\ge`	$\equiv$	`\equiv`
$\ll$	`\ll`	$\gg$	`\gg`	$\doteq$	`\doteq`
$\prec$	`\prec`	$\succ$	`\succ`	$\sim$	`\sim`
$\preceq$	`\preceq`	$\succeq$	`\succeq`	$\simeq$	`\simeq`
$\subset$	`\subset`	$\supset$	`\supset`	$\approx$	`\approx`
$\subseteq$	`\subseteq`	$\supseteq$	`\supseteq`	$\cong$	`\cong`
$\sqsubset$ [a]	`\sqsubset` [a]	$\sqsupset$ [a]	`\sqsupset` [a]	$\bowtie$	`\Join` [a]
$\sqsubseteq$	`\sqsubseteq`	$\sqsupseteq$	`\sqsupseteq`	$\bowtie$	`\bowtie`
$\in$	`\in`	$\ni$	`\ni` , `\owns`	$\propto$	`\propto`
$\vdash$	`\vdash`	$\dashv$	`\dashv`	$\models$	`\models`
$\mid$	`\mid`	$\parallel$	`\parallel`	$\perp$	`\perp`
$\smile$	`\smile`	$\frown$	`\frown`	$\asymp$	`\asymp`
$:$	`:`	$\notin$	`\notin`	$\neq$	`\neq` or `\ne`

Square Root and n^{th} Root

- The square root is entered as `\sqrt`.
- The nth root is generated with `\sqrt[`n`]`.
- The size of the root sign is determined automatically by LaTeX.
- If just the sign is needed, use `\surd`.

Square Root and n^{th} Root - Examples

```
$\sqrt{x} \Leftrightarrow x^{1/2} \quad
\sqrt[3]{2} \quad
\sqrt{x^{2} + \sqrt{y}}  \quad
\surd[x^2 + y^2]$
```

$$\sqrt{x} \Leftrightarrow x^{1/2} \quad \sqrt[3]{2} \quad \sqrt{x^{2} + \sqrt{y}} \quad \surd[x^2 + y^2]$$

Arrows

Various kinds of arrows:

$\leftarrow$	\leftarrow or \gets	$\longleftarrow$	\longleftarrow
$\rightarrow$	\rightarrow or \to	$\longrightarrow$	\longrightarrow
$\leftrightarrow$	\leftrightarrow	$\longleftrightarrow$	\longleftrightarrow
$\Leftarrow$	\Leftarrow	$\Longleftarrow$	\Longleftarrow
$\Rightarrow$	\Rightarrow	$\Longrightarrow$	\Longrightarrow
$\Leftrightarrow$	\Leftrightarrow	$\Longleftrightarrow$	\Longleftrightarrow
$\mapsto$	\mapsto	$\longmapsto$	\longmapsto
$\hookleftarrow$	\hookleftarrow	$\hookrightarrow$	\hookrightarrow
$\leftharpoonup$	\leftharpoonup	$\rightharpoonup$	\rightharpoonup
$\leftharpoondown$	\leftharpoondown	$\rightharpoondown$	\rightharpoondown
$\rightleftharpoons$	\rightleftharpoons	$\iff$	\iff (bigger spaces)
$\uparrow$	\uparrow	$\downarrow$	\downarrow
$\updownarrow$	\updownarrow	$\Uparrow$	\Uparrow
$\Downarrow$	\Downarrow	$\Updownarrow$	\Updownarrow
$\nearrow$	\nearrow	$\searrow$	\searrow
$\swarrow$	\swarrow	$\nwarrow$	\nwarrow
$\leadsto$	\leadsto [a]		

Dots

- Use `\cdot` to typeset a single centered dot.
- `\cdots` is three centered dots while `\ldots` sets the dots low (on the baseline).
- Besides that, there are `\vdots` for vertical and `\ddots` for diagonal dots.

Dots - Examples

```
$\Psi = v_1 \cdot v_2 \cdot \ldots \qquad
n! = 1 \cdot 2 \cdots (n-1) \cdot n$
```

$$\Psi = v_1 \cdot v_2 \cdot \ldots \qquad n! = 1 \cdot 2 \cdots (n-1) \cdot n$$

Horizontal Lines

The commands \overline and \underline create horizontal lines directly over or under an expression:

```
$0.\overline{3} =
\underline{\underline{1/3}}$
```

$$0.\overline{3} = \underline{\underline{1/3}}$$

Horizontal Braces

The commands \overbrace and \underbrace create long horizontal braces over or under an expression:

```
$\underbrace{\overbrace{a+b+c}^6 \cdot
\overbrace{d+e+f}^7}_\text{meaning of
life} = 42$
```

$$\underbrace{\overbrace{a+b+c}^{6} \cdot \overbrace{d+e+f}^{7}}_{\text{meaning of life}} = 42$$

Wide Hats, Tildes and Primes

- Wide hats and tildes covering several characters are generated with `\widetilde` and `\widehat`.
- Notice the difference between `\hat` and `\widehat` and the placement of `\bar` for a variable with subscript.
- The apostrophe mark `'` gives a prime.

Wide Hats, Tildes and Primes - Examples

```
$f(x) = x^2 \qquad f'(x) = 2x \qquad
f''(x) = 2$\\[5pt]
$\hat{XY} \quad \widehat{XY} \quad
\bar{x_0} \quad \bar{x}_0$
```

$$f(x) = x^2 \qquad f'(x) = 2x \qquad f''(x) = 2$$

$$\hat{XY} \quad \widehat{XY} \quad \bar{x_0} \quad \bar{x}_0$$

Math Mode Accents

To add mathematical accents such as small arrows or tilde signs to variables use commands:

$\hat{a}$	\hat{a}	$\check{a}$	\check{a}	$\tilde{a}$	\tilde{a}
$\grave{a}$	\grave{a}	$\dot{a}$	\dot{a}	$\ddot{a}$	\ddot{a}
$\bar{a}$	\bar{a}	$\vec{a}$	\vec{a}	$\widehat{AAA}$	\widehat{AAA}
$\acute{a}$	\acute{a}	$\breve{a}$	\breve{a}	$\widetilde{AAA}$	\widetilde{AAA}
$\mathring{a}$	\mathring{a}				

Vectors

- Vectors are specified with the `\vec` command.
- The two commands `\overrightarrow` and `\overleftarrow` are useful to denote the vector from A to B:

```
$\vec{a} \qquad \vec{AB} \qquad
\overrightarrow{AB}$
```

$$\vec{a} \qquad \vec{AB} \qquad \overrightarrow{AB}$$

Function Names

LaTeX supplies the following commands to typeset the most common function names:

```
\arccos \cos  \csc \exp  \ker    \limsup
\arcsin \cosh \deg \gcd  \lg     \ln
\arctan \cot  \det \hom  \lim    \log
\arg    \coth \dim \inf  \liminf \max
\sinh   \sup  \tan \tanh \min    \Pr
\sec    \sin
```

Function Names - Example

```
\begin{equation*} \lim_{x \rightarrow 0}
\frac{\sin x}{x}=1  \end{equation*}
```

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}=1$$

Modulo Function

For the modulo function, there are two commands `\bmod` and `\pmod` :

```
$a\bmod b$ \\
$x\equiv a \pmod{b}$
```

$a \bmod b$
$x \equiv a \pmod{b}$

Fractions

- A built-up fraction is typeset with the `\frac{...}{...}` command.
- In in-line equations, the fraction is shrunk to fit the line.
- This style is obtainable in display style with `\tfrac`.
- The reverse, i.e. display style fraction in text, is made with `\dfrac`.

Fractions - Examples

```
In display style:
\begin{equation*}
3/8 \qquad \frac{3}{8} \qquad \tfrac{3}{8}
\end{equation*}
```

In display style:

$$3/8 \qquad \frac{3}{8} \qquad \tfrac{3}{8}$$

Fractions - Examples

```
In text style:
$1\frac{1}{2}$~hours \qquad
$1\dfrac{1}{2}$~hours
```

In text style: $1\frac{1}{2}$ hours $\qquad$ $1\dfrac{1}{2}$ hours

Partial Derivatives

The \partial command for partial derivatives:

```
\begin{equation*}
\sqrt{\frac{x^2}{k+1}}\qquad
x^\frac{2}{k+1}\qquad
\frac{\partial^2f}{\partial x^2}
\end{equation*}
```

$$\sqrt{\frac{x^2}{k+1}} \qquad x^{\frac{2}{k+1}} \qquad \frac{\partial^2 f}{\partial x^2}$$

Binomial Coefficients

For binomial coefficients or similar structures, use the command `\binom`:

```
\begin{equation*}
\binom{n}{k} = \binom{n-1}{k} +
\binom{n-1}{k-1}
\end{equation*}
```

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

Stacking Symbols

- For binary relations it may be useful to stack symbols over each other.
- `\stackrel{#1}{#2}` puts the symbol given in #1 in superscript-like size over #2 which is set in its usual position.

```
\begin{equation*}
f_n(x) \stackrel{*}{\approx} 1
\end{equation*}
```

$$f_n(x) \stackrel{*}{\approx} 1$$

Integral, Sum, Product Operators

- The integral operator is generated with `\int`.
- The sum operator is generated with `\sum`.
- The product operator is generated with `\prod`.

Integral, Sum, Product Operators-Examples

```
\begin{equation*}
\sum_{i=1}^n \qquad
\int_0^{\frac{\pi}{2}} \qquad
\prod_\epsilon
\end{equation*}
```

$$\sum_{i=1}^{n} \qquad \int_{0}^{\frac{\pi}{2}} \qquad \prod_{\epsilon}$$

Complex Expressions

To get more control over the placement of indices in complex expressions use the `\substack` command:

```
\begin{equation*}
\sum^n_{\substack{0<i<n \\ j\subseteq i}}
P(i,j) = Q(i,j)
\end{equation*}
```

$$\sum^n_{\substack{0<i<n \\ j\subseteq i}} P(i,j) = Q(i,j)$$

Brackets and Delimiters

- LaTeX provides all sorts of symbols for bracketing and other delimiters (e.g. $[$ $\langle$ $\|$ $\updownarrow$).
- Round and square brackets can be entered with the corresponding keys and curly braces with `\{`, but all other delimiters are generated with special commands (e.g. `\updownarrow`).

Brackets and Delimiters - Examples

```
\begin{equation*}
{a,b,c} \neq \{a,b,c\}
\end{equation*}
```

$$a, b, c \neq \{a, b, c\}$$

Brackets and Delimiters

- If you put `\left` in front of an opening delimiter and `\right` in front of a closing delimiter, LaTeX will automatically determine the correct size of the delimiter.
- Note that you must close every `\left` with a corresponding `\right`.
- If you don't want anything on the right, use the "invisible `\right.`":

Brackets and Delimiters - Examples

```
\begin{equation*}
1 + \left(\frac{1}{1-x^{2}} \right)^3
\qquad \left. \ddagger \frac{~}{~}\right)
\end{equation*}
```

$$1 + \left(\frac{1}{1-x^{2}} \right)^3 \qquad \left. \ddagger \frac{~}{~}\right)$$

Brackets and Delimiters

In some cases it is necessary to specify the correct size of a mathematical delimiter by hand, which can be done using the commands

`\big`, `\Big`, `\bigg` and `\Bigg`

as prefixes to most delimiter commands, but try to avoid this.

Brackets and Delimiters - Examples

```
$\Big((x+1)(x-1)\Big)^{2}$\\
$\big( \Big( \bigg( \Bigg( \quad
\big\} \Big\} \bigg\} \Bigg\} \quad
\big\| \Big\| \bigg\| \Bigg\| \quad
\big\Downarrow \Big\Downarrow
\bigg\Downarrow \Bigg\Downarrow$
```

$$\Big((x+1)(x-1)\Big)^{2}$$

$$\big(\Big(\bigg(\Bigg(\quad \big\} \Big\} \bigg\} \Bigg\} \quad \big\| \Big\| \bigg\| \Bigg\| \quad \big\Downarrow \Big\Downarrow \bigg\Downarrow \Bigg\Downarrow$$

Delimiters

A list of all delimiters available:

(	(	)	)	↑	\uparrow
[	[or \lbrack	]	] or \rbrack	↓	\downarrow
{	\{ or \lbrace	}	\} or \rbrace	↕	\updownarrow
⟨	\langle	⟩	\rangle	⇑	\Uparrow
|	| or \vert	‖	\\| or \Vert	⇓	\Downarrow
/	/	\	\backslash	⇕	\Updownarrow
⌊	\lfloor	⌋	\rfloor		
⌉	\rceil	⌈	\lceil		

Single Equations that are Too Long and Multiple Equations

Single Equations that are Too Long and Multiple Equations

If an equation is too long, we have to wrap it somehow and we should follow some rules:

- In general always wrap an equation before an equality sign or an operator.
- A wrap before an equality sign is preferable to a wrap before any operator.
- A wrap before a plus- or minus-operator is preferable to a wrap before a multiplication-operator.
- Any other type of wrap should be avoided if at all possible.

Multline Environment

The easiest way to achieve such a wrapping is the use of the **multline** environment:

```
\begin{multline}
a + b + c + d + e + f + g + h + i + j + k
+ l + m + n + o \\ = j + k + l + m + n
\end{multline}
```

$$\begin{multline} a+b+c+d+e+f+g+h+i+j+k+l+m+n+o \\ = j+k+l+m+n \end{multline} \tag{4}$$

Multline Environment

- The main difference between **multline** environment and the **equation** environment is that an arbitrary line-break (or also multiple line-breaks) can be introduced.
- Similarly to **equation*** there also exists a **multline*** version.
- A better solution is offered by the **eqnarray** (**eqnarray***) environment.

Eqnarray Environment - Example

```
\begin{eqnarray}
a & = & b + c \\
  & = & d + e + f + g + h + i + j + k + l \\
  &   & + m + n + o \\
  & = & p + q + r + s
\end{eqnarray}
```

$$
\begin{aligned}
a &= b + c && (5)\\
&= d + e + f + g + h + i + j + k + l && (6)\\
&\quad + m + n + o && (7)\\
&= p + q + r + s && (8)
\end{aligned}
$$

Eqnarray Environment - Example

```
\begin{eqnarray}
a & = & b + c \\
  & = & d + e + f + g + h + i + j + k + l \\
  &   & + m + n + o \nonumber \\
  & = & p + q + r + s
\end{eqnarray}
```

$$
\begin{aligned}
a &= b + c && (9)\\
&= d + e + f + g + h + i + j + k + l && (10)\\
&\quad + m + n + o \\
&= p + q + r + s && (11)
\end{aligned}
$$

Eqnarray* Environment - Example

```
\begin{eqnarray*}
a & = & b + c \\
  & = & d + e + f + g + h + i + j + k + l \\
  &   & + m + n + o \\
  & = & p + q + r + s
\end{eqnarray*}
```

$$
\begin{aligned}
a &= b + c \\
&= d + e + f + g + h + i + j + k + l \\
&\quad + m + n + o \\
&= p + q + r + s
\end{aligned}
$$

Arrays and Matrices

Array and Cases Environment

- To typeset arrays (e.g. math tables), use the **array** environment.
- The `\\` command is used to break lines.
- The **array** environment can also be used to typeset piecewise functions by using a `\right.` as an invisible right delimiter.
- The **cases** environment from **amsmath** package simplifies the syntax for typesetting piecewise functions.

Array - Example

```
\begin{equation*}
\mathbf{X} = \left(
\begin{array}{ccc}
x_1 & x_2 & \ldots \\
x_3 & x_4 & \ldots \\
\vdots & \vdots & \ddots
\end{array} \right)
\end{equation*}
```

$$\mathbf{X} = \left(\begin{array}{ccc} x_1 & x_2 & \ldots \\ x_3 & x_4 & \ldots \\ \vdots & \vdots & \ddots \end{array} \right)$$

Array - Example

```
\begin{equation*}
|x| = \left\{
\begin{array}{rl}
-x & \text{if } x < 0,\\
0 & \text{if } x = 0,\\
x & \text{if } x > 0.
\end{array} \right.
\end{equation*}
```

$$|x| = \left\{ \begin{array}{rl} -x & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ x & \text{if } x > 0. \end{array} \right.$$

Cases - Example

```
\begin{equation*}
|x| =
\begin{cases}
-x & \text{if } x < 0,\\
0 & \text{if } x = 0,\\
x & \text{if } x > 0.
\end{cases}
\end{equation*}
```

$$|x| = \begin{cases} -x & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ x & \text{if } x > 0. \end{cases}$$

Matrices

- Matrices can be typeset by **array**, but **amsmath** package provides a better solution using the different **matrix** environments.
- There are six versions with different delimiters: **matrix** (none), **pmatrix** (, **bmatrix** [, **Bmatrix** {, **vmatrix** | and **Vmatrix** ‖.
- You don't have to specify the number of columns as with **array**.
- The maximum number is 10, but it is customisable.

Matrices - Examples

```
\begin{equation*}\begin{matrix}
1 & 2 \\
3 & 4
\end{matrix} \qquad
\begin{bmatrix}
p_{11} & p_{12} & \ldots & p_{1n} \\
\vdots & \vdots & \ddots & \vdots \\
p_{m1} & p_{m2} & \ldots & p_{mn}
\end{bmatrix}\end{equation*}
```

$$\begin{matrix} 1 & 2 \\ 3 & 4 \end{matrix} \qquad \begin{bmatrix} p_{11} & p_{12} & \ldots & p_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \ldots & p_{mn} \end{bmatrix}$$

Spacing in Math Mode

Spacing in Math Mode

Spacing in math mode can be adjusted by inserting special spacing commands:

- `\␣` generates a medium sized space comparable to the interword spacing.
- The size of a `\quad` corresponds to the width of the character 'M' of the current font, and `\qquad` produce even larger space.
- Some other spacing commands: `\,` for $\frac{3}{18}\,\mathrm{quad}$, `\:` for $\frac{4}{18}\,\mathrm{quad}$ and `\;` for $\frac{5}{18}\,\mathrm{quad}$.
- `\!` produces a negative space of $-\frac{3}{18}\,\mathrm{quad}$.

Spacing in Math Mode - Examples

```
\begin{equation*}
\int_1^2 \ln x \mathrm{d}x
\qquad
\int_1^2 \ln x \,\mathrm{d}x
\end{equation*}
```

$$\int_1^2 \ln x \mathrm{d}x \qquad \int_1^2 \ln x \,\mathrm{d}x$$

Multiple Integrals

- If you want to typeset multiple integrals, you'll discover that the spacing between the integrals is too wide.
- You can correct it using `\!`, but **amsmath** provides an easier way for fine-tuning the spacing, namely the `\iint`, `\iiint` and `\iiiint` commands.

Multiple Integrals - Examples

```
\begin{eqnarray*}
\int\int f(x)g(y) \, \mathrm{d}x\mathrm{d}y\\
\int\!\!\!\int f(x)g(y) \, \mathrm{d}x\mathrm{d}y\\
\iint f(x)g(y) \, \mathrm{d}x\mathrm{d}y
\end{eqnarray*}
```

$$\int\int f(x)g(y)\,\mathrm{d}x\mathrm{d}y$$

$$\int\!\!\!\int f(x)g(y)\,\mathrm{d}x\mathrm{d}y$$

$$\iint f(x)g(y)\,\mathrm{d}x\mathrm{d}y$$

Phantoms

Using the \phantom command you can reserve space for characters that do not show up in the final output:

```
\begin{equation*}
{}^{14}_{6}\text{C} \qquad \text{versus}
\qquad {}^{14}_{\phantom{1}6}\text{C}
\end{equation*}
```

$$ {}^{14}_{6}\text{C} \qquad \text{versus} \qquad {}^{14}_{6}\text{C} $$

Math Fonts and Alphabets

Math Fonts

`\mathrm{...}`	Roman Font
`\mathbf{...}`	**Boldface Font**
`\mathsf{...}`	Sans Serif Font
`\mathtt{...}`	Typewriter Font
`\mathit{...}`	*Italic Font*
`\mathcal{...}`	$\mathcal{CALLIGRAPHIC\ FONT}$
`\mathnormal{...}`	*Normal Font*

Math Alphabets

Example	Command	Required package
ABCDEabcde1234	`\mathrm{ABCDE abcde 1234}`	
$ABCDEabcde1234$	`\mathit{ABCDE abcde 1234}`	
$ABCDEabcde1234$	`\mathnormal{ABCDE abcde 1234}`	
$\mathcal{ABCDE}$	`\mathcal{ABCDE abcde 1234}`	
$\mathscr{ABCDE}$	`\mathscr{ABCDE abcde 1234}`	mathrsfs
$\mathfrak{ABCDEabcde1234}$	`\mathfrak{ABCDE abcde 1234}`	amsfonts or amssymb
$\mathbb{ABCDE}\nexists\nexists\nexists\nexists$	`\mathbb{ABCDE abcde 1234}`	amsfonts or amssymb

Math Fonts and Alphabets - Examples

```
$\mathscr{R} \qquad
 \mathcal{R}  \qquad
 \mathfrak{R} \qquad
 \mathbb{R}   \qquad$
```

$\mathscr{R} \qquad \mathcal{R} \qquad \mathfrak{R} \qquad \mathbb{R}$

The first requires **mathrsfs** package and the last two require **amssymb** or **amsfonts** package.

Font Size

- Sometimes you need to tell LaTeX the correct font size.
- In math mode, this is set with the following four commands:

  `\displaystyle` (123),
  `\textstyle` (123),
  `\scriptstyle` (123) and
  `\scriptscriptstyle` (123).

Font Size - Examples

```
\begin{equation*}
P = \frac{\displaystyle{\sum_{i=1}^n (x_i- x)
(y_i- y)}}{\displaystyle{\left[\sum_{i=1}^n
(x_i-x)^2\sum_{i=1}^n(y_i- y)^2\right]^{1/2}}}
\end{equation*}
```

$$P = \frac{\displaystyle{\sum_{i=1}^n (x_i- x)(y_i- y)}}{\displaystyle{\left[\sum_{i=1}^n (x_i-x)^2\sum_{i=1}^n(y_i- y)^2\right]^{1/2}}}$$

Bold Symbols

The package **amsbsy** (included by **amsmath**) includes a \boldsymbol command:

```
$\mu, \qquad \boldsymbol{\mu}$
```

$\mu, \qquad \boldsymbol{\mu}$

Theorems, Lemmas, Definitions, . . .

Theorems, . . .

- When writing mathematical documents, you probably need a way to typeset "Lemmas", "Definitions", "Axioms" and similar structures.
- Use command:

```
\newtheorem{name}[counter]{text}[section]
```

Theorems, . . .

```
\newtheorem{name}[counter]{text}[section]
```

- *name* is a short keyword used to identify the "theorem".
- *text* defines the actual name of the "theorem", which will be printed in the final document.
- *counter* specifies the *name* of a previously declared "theorem".
- The new "theorem" will then be numbered in the same sequence.
- *section* specifies the sectional unit within which the "theorem" should get its numbers.

Theorems, . . .

After executing the \newtheorem command in the preamble of your document, you can use the following command within the document:

```
\begin{name}
  This is my interesting theorem
\end{name}
```

Theorems, . . . - Example

```
%Preamble:
\newtheorem{mur}{Murphy}

%Body:
\begin{mur}
If there are two or more ways to do
something, and one of those ways can
result in a catastrophe, then someone
will do it.
\end{mur}
```

Proofs

The **amsthm** package also provides the **proof** environment:

```
\begin{proof}
Trivial, use
\begin{equation*}
E=mc^2.
\end{equation*}
\end{proof}
```

Proof.

Trivial, use

$$E = mc^2.$$

□

Proofs

With the command `\qedhere` you can move the 'end of proof' symbol around for situations where it would end up alone on a line.

```
\begin{proof}
Trivial, use
\begin{equation*}
E=mc^2. \qedhere
\end{equation*}
\end{proof}
```

Proof.

Trivial, use

$$E = mc^2. \qedhere$$

List of Mathematical Symbols

Binary Operators

$+$	`+`	$-$	`-`		
$\pm$	`\pm`	$\mp$	`\mp`	$\triangleleft$	`\triangleleft`
$\cdot$	`\cdot`	$\div$	`\div`	$\triangleright$	`\triangleright`
$\times$	`\times`	$\setminus$	`\setminus`	$\star$	`\star`
$\cup$	`\cup`	$\cap$	`\cap`	$*$	`\ast`
$\sqcup$	`\sqcup`	$\sqcap$	`\sqcap`	$\circ$	`\circ`
$\vee$	`\vee , \lor`	$\wedge$	`\wedge , \land`	$\bullet$	`\bullet`
$\oplus$	`\oplus`	$\ominus$	`\ominus`	$\diamond$	`\diamond`
$\odot$	`\odot`	$\oslash$	`\oslash`	$\uplus$	`\uplus`
$\otimes$	`\otimes`	$\bigcirc$	`\bigcirc`	$\amalg$	`\amalg`
$\bigtriangleup$	`\bigtriangleup`	$\bigtriangledown$	`\bigtriangledown`	$\dagger$	`\dagger`
$\lhd$ [a]	`\lhd` [a]	$\rhd$ [a]	`\rhd` [a]	$\ddagger$	`\ddagger`
$\unlhd$ [a]	`\unlhd` [a]	$\unrhd$ [a]	`\unrhd` [a]	$\wr$	`\wr`

BIG Operators

$\sum$	\sum	$\bigcup$	\bigcup	$\bigvee$	\bigvee
$\prod$	\prod	$\bigcap$	\bigcap	$\bigwedge$	\bigwedge
$\coprod$	\coprod	$\bigsqcup$	\bigsqcup	$\biguplus$	\biguplus
$\int$	\int	$\oint$	\oint	$\bigodot$	\bigodot
$\bigoplus$	\bigoplus	$\bigotimes$	\bigotimes		

Arrows as Accents and Large Delimiters

$\overrightarrow{AB}$	`\overrightarrow{AB}`	$\underrightarrow{AB}$	`\underrightarrow{AB}`
$\overleftarrow{AB}$	`\overleftarrow{AB}`	$\underleftarrow{AB}$	`\underleftarrow{AB}`
$\overleftrightarrow{AB}$	`\overleftrightarrow{AB}`	$\underleftrightarrow{AB}$	`\underleftrightarrow{AB}`

$\lgroup$	`\lgroup`	$\rgroup$	`\rgroup`	$\lmoustache$	`\lmoustache`
$\arrowvert$	`\arrowvert`	$\Arrowvert$	`\Arrowvert`	$\bracevert$	`\bracevert`
$\rmoustache$	`\rmoustache`				

Miscellaneous Symbols

$\dots$	\dots	$\cdots$	\cdots	$\vdots$	\vdots	$\ddots$	\ddots
$\hbar$	\hbar	$\imath$	\imath	$\jmath$	\jmath	ℓ	\ell
$\Re$	\Re	$\Im$	\Im	$\aleph$	\aleph	$\wp$	\wp
$\forall$	\forall	$\exists$	\exists	$\mho$ [a]	\mho [a]	∂	\partial
$'$	'	$\prime$	\prime	$\emptyset$	\emptyset	∞	\infty
∇	\nabla	$\triangle$	\triangle	$\square$	\Box [a]	$\diamond$	\Diamond [a]
$\bot$	\bot	$\top$	\top	$\angle$	\angle	$\surd$	\surd
$\diamondsuit$	\diamondsuit	$\heartsuit$	\heartsuit	$\clubsuit$	\clubsuit	$\spadesuit$	\spadesuit
$\neg$	\neg or \lnot	$\flat$	\flat	$\natural$	\natural	$\sharp$	\sharp